

FLOOD SUSCEPTIBILITY MAPPING OF DELHI USING GIS AND MULTI-CRITERIA DECISION MAKING (MCDM)

Abstract

Flooding is one of the most frequent and destructive natural hazards affecting urban environments. Delhi experiences recurrent flood and waterlogging events due to rapid urbanization, increasing impervious surfaces, encroachment on floodplains, and the influence of the Yamuna River. This study aims to identify flood-prone areas within Delhi using Geographic Information Systems (GIS), Remote Sensing, and Multi-Criteria Decision Making (MCDM) techniques. Five flood conditioning factors, namely elevation, slope, distance from river, land use/land cover (LULC), and rainfall, were considered for flood susceptibility assessment. SRTM Digital Elevation Model (DEM) data were used to derive elevation and slope layers, while proximity analysis was used to generate the distance-to-river layer. Each factor was reclassified according to its contribution to flooding and integrated using the Weighted Linear Combination (WLC) approach. The assigned weights were 30% for distance from river, 25% for elevation, 20% for LULC, 13% for rainfall, and 12% for slope. The final flood susceptibility map classified Delhi into Very Low, Low, Moderate, High, and Very High susceptibility zones. The results indicate that areas adjacent to the Yamuna River and low-lying eastern regions of Delhi exhibit the highest flood susceptibility, while western and southwestern regions show comparatively lower susceptibility. The generated flood susceptibility map can support disaster management agencies, urban planners, and policymakers in flood mitigation and sustainable urban development.

Keywords: Flood Susceptibility, GIS, MCDM, Weighted Overlay Analysis, Remote Sensing, Delhi, Yamuna River.

CHAPTER 1: INTRODUCTION

1.1 Background

Floods are among the most devastating natural hazards worldwide, causing significant loss of life, infrastructure damage, and economic disruption. Urban flooding has become increasingly severe due to rapid urbanization, inadequate drainage infrastructure, and encroachment upon natural floodplains. In India, major metropolitan cities frequently experience urban flooding during monsoon seasons.

Delhi, the National Capital Territory of India, is highly vulnerable to flooding because of its proximity to the Yamuna River and extensive urban development within flood-prone regions. Historical flood events have repeatedly affected areas situated along the Yamuna floodplain. Studies have identified the Yamuna corridor as one of the most vulnerable regions within Delhi due to its low elevation and proximity to the river.

Geographic Information Systems (GIS) and Remote Sensing provide efficient tools for flood susceptibility assessment through the integration of multiple environmental factors. Multi-Criteria Decision Making (MCDM) approaches enable the evaluation and weighting of these factors to generate flood susceptibility maps.

1.2 Problem Statement

Rapid urban expansion, increasing built-up areas, and encroachment on floodplains have increased flood vulnerability in Delhi. Identifying flood-prone areas is essential for effective flood management, urban planning, and disaster preparedness.

1.3 Objectives

1. To generate flood conditioning factor maps for Delhi.
 2. To apply GIS-based Multi-Criteria Decision Making (MCDM) techniques.
 3. To perform Weighted Overlay Analysis for flood susceptibility assessment.
 4. To identify areas vulnerable to flooding.
 5. To prepare a flood susceptibility map of Delhi.
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1.4 Study Area

Delhi is located in northern India between approximately 28.40°N–28.88°N latitude and 76.84°E–77.35°E longitude.

Characteristics of the study area include:

- Area: Approximately 1484 km²
- Major River: Yamuna River
- Climate: Semi-arid to subtropical
- Average Annual Rainfall: 600–800 mm
- Elevation Range: Approximately 200–320 m above mean sea level

The Yamuna River forms the primary drainage system of Delhi and significantly influences flood occurrence within the city. Historical flooding has repeatedly affected settlements along the Yamuna floodplain.

CHAPTER 2: LITERATURE REVIEW

Tomar et al. (2021) conducted a GIS-based urban flood risk assessment for Delhi and demonstrated that the Yamuna River floodplain represents one of the most flood-prone regions within the National Capital Territory. Their study integrated hydrological and spatial datasets to identify flood-affected zones.

Khan (2014) examined the geomorphology and floodplain characteristics of the Yamuna River in Delhi and highlighted the reduction of natural floodplain areas due to urban encroachment. The study emphasized the significance of preserving floodplains for flood mitigation.

Recent flood mapping studies based on the 2023 Delhi floods have shown that flood extent was concentrated along the Yamuna corridor and low-lying eastern regions of Delhi. Satellite-based flood monitoring confirmed substantial flood inundation during the peak flooding period.

Several studies have successfully employed GIS-based Weighted Overlay Analysis and Multi-Criteria Decision-Making techniques for flood susceptibility mapping by integrating elevation, slope, rainfall, proximity to rivers, and land use characteristics.

CHAPTER 3: DATA USED

3.1 Data Collection

The datasets used for flood susceptibility mapping of Delhi were obtained from various open-source and publicly available platforms. These datasets were processed in QGIS for generating flood conditioning factors and conducting Weighted Overlay Analysis.

Table 3.1 Data Sources Used in the Study

S.No.	Factor (Xi)	Data Source
1	Distance from River	OpenStreetMap (OSM)
2	Elevation	SRTM DEM 30 m
3	Land Use/Land Cover (LULC)	ESA WorldCover 2021
4	Rainfall	CHIRPS Dataset
5	Slope	Derived from SRTM DEM 30 m
	Total Factors Used	5

3.2 Description of Data Sources

Distance from River

The Yamuna River layer was extracted from OpenStreetMap (OSM) and used to generate the distance-from-river raster through proximity analysis. Areas closer to the river were considered more susceptible to flooding.

Elevation

Elevation data were obtained from the Shuttle Radar Topography Mission (SRTM) Digital Elevation Model with a spatial resolution of 30 meters. The DEM was clipped to the Delhi boundary and used for elevation analysis.

Land Use/Land Cover (LULC)

LULC data were obtained from the ESA WorldCover 2021 dataset and classified into major land cover categories including built-up area, vegetation, water bodies, and barren land.

Rainfall

Rainfall information was obtained from the Climate Hazards Group InfraRed Precipitation with Stations (CHIRPS) dataset and used to represent the spatial distribution of rainfall across the study area.

Slope

The slope layer was derived from the SRTM DEM using terrain analysis tools available in QGIS. Areas with lower slopes were assigned higher flood susceptibility scores.

CHAPTER 4: METHODOLOGY

4.1 Methodological Framework

Data Collection

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Data Pre-processing

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Generation of Flood Conditioning Factors

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Reclassification

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Weight Assignment

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Weighted Overlay Analysis

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Flood Susceptibility Mapping

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Result Interpretation

4.2 Preparation of Flood Conditioning Factors

4.2.1 Elevation

Elevation influences the accumulation and movement of surface water. Lower elevation areas are generally more susceptible to flooding than higher areas. Elevation was derived from SRTM DEM data and reclassified into susceptibility classes.

4.2.2 Slope

Slope controls runoff velocity and water accumulation. Areas with lower slopes tend to accumulate water and therefore exhibit higher flood susceptibility. Slope was generated from the DEM using terrain analysis tools in QGIS.

4.2.3 Distance from River

Distance from the Yamuna River is one of the most important flood conditioning factors in Delhi. Areas located closer to the river have a greater probability of experiencing flood inundation. A proximity analysis was performed to generate a distance-to-river raster.

4.2.4 Land Use/Land Cover (LULC)

Land use influences infiltration and runoff generation. Built-up areas generally exhibit higher runoff and lower infiltration compared to vegetated surfaces. The LULC layer was classified into built-up, vegetation, water bodies, and barren land categories.

4.2.5 Rainfall

Rainfall is a primary triggering factor for flooding. Areas receiving higher rainfall intensities are generally more vulnerable to flood occurrence. Rainfall data were processed and reclassified into flood susceptibility classes.

4.3 Drainage Density Analysis

Drainage density was initially considered as a flood conditioning factor.

Hydrological analysis was performed using SRTM DEM data through flow direction, flow accumulation, stream extraction, and stream network generation techniques. Additionally, line density analysis using OpenStreetMap waterway data was conducted.

However, due to the relatively flat terrain of Delhi and limitations associated with the 30-meter resolution SRTM DEM, the generated drainage networks exhibited unrealistic drainage patterns and inconsistent density distributions. The resulting drainage density layers did not accurately represent the hydrological characteristics of the study area.

Therefore, drainage density was excluded from the final flood susceptibility model to avoid introducing uncertainty into the analysis.

4.4 Multi-Criteria Decision Making (MCDM)

Flood occurrence is controlled by multiple interacting environmental factors. Therefore, a Multi-Criteria Decision Making (MCDM) approach was adopted for integrating different flood conditioning factors.

The Weighted Linear Combination (WLC) method was selected due to its simplicity, transparency, and widespread application in flood susceptibility studies.

4.5 Weighted Overlay Analysis

The Weighted Overlay Analysis was performed by assigning weights to each flood conditioning factor based on its relative importance.

The Flood Susceptibility Index (FSI) was calculated using:

$$FSI = \sum W_i X_i \quad (i = 1 \text{ to } n)$$

Where:

- FSI = Flood Susceptibility Index
- W_i = Weight of factor i

- X_i = Reclassified score of factor i

The weighted overlay equation used in this study was:

$$\text{Flood Susceptibility Index (FSI)} = (0.30 \times \text{Distance from River}) + (0.25 \times \text{Elevation}) + (0.20 \times \text{LULC}) + (0.13 \times \text{Rainfall}) + (0.12 \times \text{Slope})$$

4.6 Weight Assignment

Factor	Weight (%)
Distance from River	30
Elevation	25
LULC	20
Rainfall	13
Slope	12
Total	100

The weights were assigned based on literature review and the relative influence of each factor on flood occurrence.

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Elevation Analysis

The elevation map indicates that eastern Delhi exhibits relatively lower elevations compared to western and southwestern regions. Lower elevation areas facilitate water accumulation and are therefore more susceptible to flooding.

5.2 Slope Analysis

The slope map reveals that a large portion of Delhi is characterized by gentle slopes. Such areas promote water stagnation and increase flood susceptibility.

5.3 Distance from River Analysis

The distance-to-river analysis highlights the Yamuna floodplain as the most vulnerable region. Flood susceptibility decreases progressively with increasing distance from the river.

5.4 Rainfall Analysis

Rainfall distribution exhibits spatial variability across Delhi. Areas receiving higher rainfall were assigned greater flood susceptibility scores.

5.5 LULC Analysis

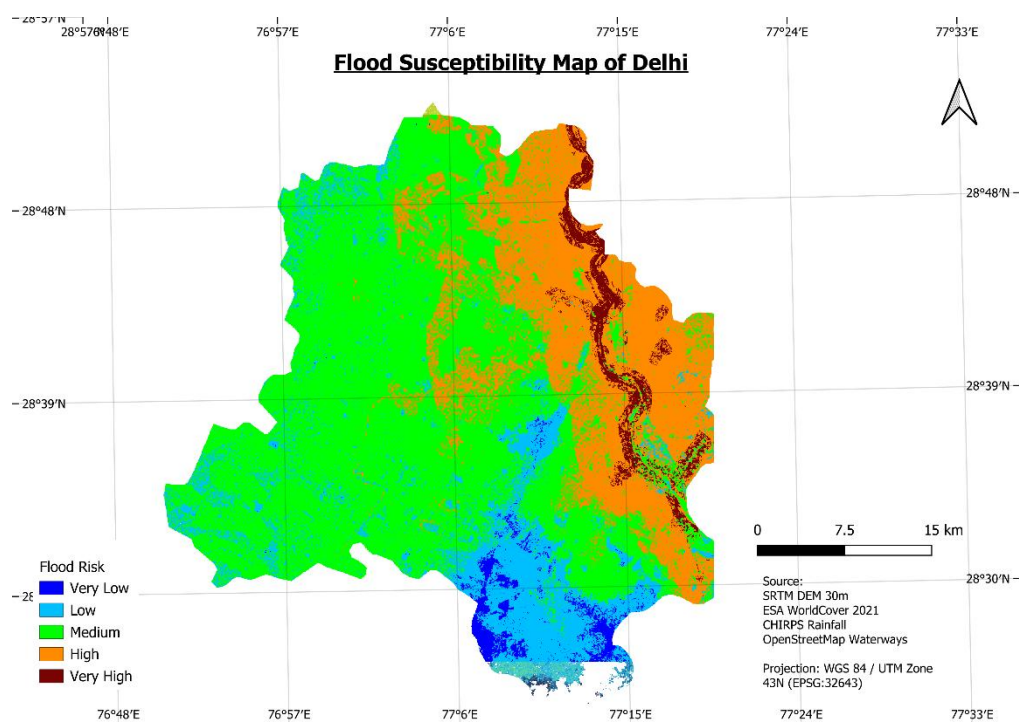
Built-up areas were identified as highly susceptible due to increased impervious surfaces and reduced infiltration.

5.6 Final Flood Susceptibility Map

The final flood susceptibility map was generated using the Weighted Overlay Analysis based on five flood conditioning factors. Areas adjacent to the Yamuna River exhibit Very High flood susceptibility, whereas western and southwestern regions generally fall under Low to Moderate susceptibility categories.

The final flood susceptibility map classified Delhi into five categories:

Class	Flood Susceptibility
1	Very Low
2	Low
3	Moderate
4	High
5	Very High



5.7 Validation and Comparison with Observed Flood Conditions

The generated flood susceptibility map was visually compared with historically flood-prone regions reported along the Yamuna floodplain. The high susceptibility zones identified in this study correspond closely with flood-prone areas documented during recent flooding events in Delhi, indicating the reliability of the adopted methodology.

The analysis indicates that areas located along the Yamuna floodplain exhibit High to Very High flood susceptibility. Moderate susceptibility dominates central Delhi, while western and southwestern regions generally exhibit Low to Very Low susceptibility.

The results are consistent with documented flood-prone areas and floodplain regions identified in previous studies and official flood management assessments.

CHAPTER 6: CONCLUSION

This study successfully developed a GIS-based flood susceptibility map of Delhi using Multi-Criteria Decision Making (MCDM) and Weighted Overlay Analysis. Five flood conditioning factors including elevation, slope, distance from river, land use/land cover, and rainfall were integrated to generate the Flood Susceptibility Index.

The results indicate that the highest flood susceptibility occurs along the Yamuna River floodplain and low-lying eastern regions of Delhi. Western and southwestern regions exhibit comparatively lower susceptibility due to higher elevations and greater distance from the river.

The generated flood susceptibility map can support urban planning, flood mitigation strategies, and disaster preparedness initiatives. The methodology can also be applied to other urban areas facing flood-related challenges.

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